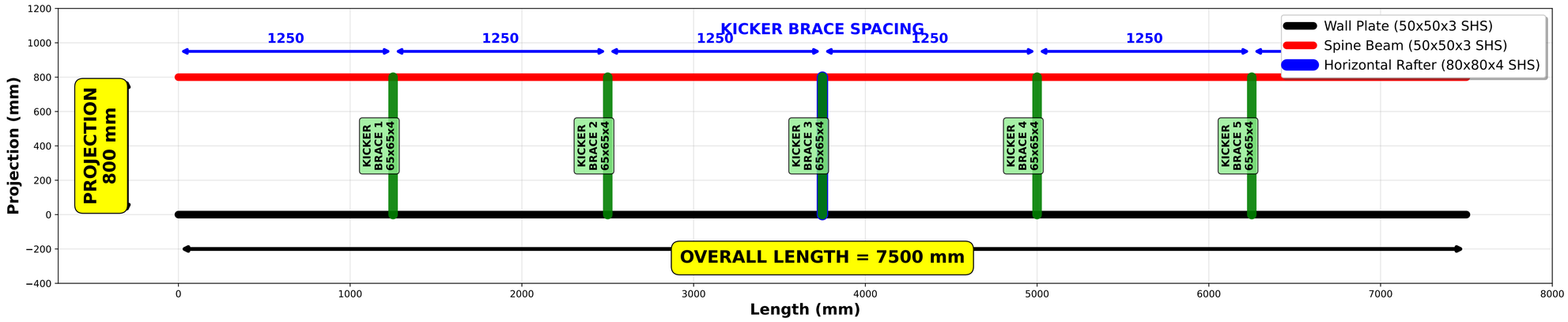
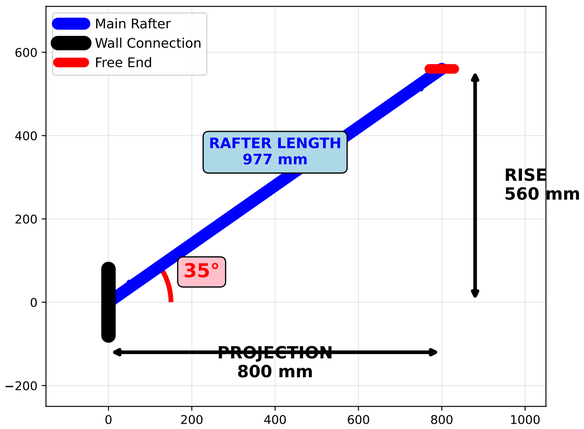
CANTILEVERED MONOPITCH AWNING - COMPLETE STRUCTURAL DESIGN
Length: 7500mm | Projection: 800mm | Slope: 35°

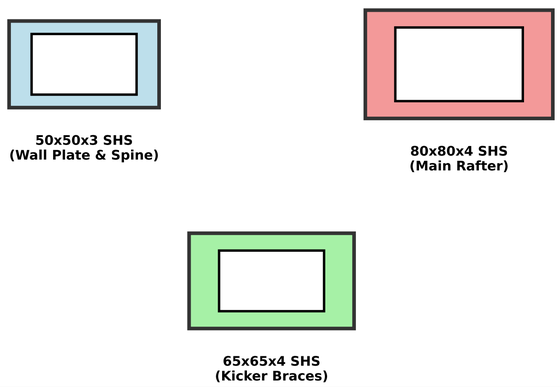
STRUCTURAL PLAN - CANTILEVERED MONOPITCH AWNING



SIDE ELEVATION



SECTION PROFILES



STRUCTURAL ANALYSIS SUMMARY

Member	Size	Stress (MPa)	Utilization	Status
Wall Plate	50x50x3 SHS	189	68.6%	✓ ADEQUATE
Spine Beam	50x50x3 SHS	189	68.6%	✓ ADEQUATE
Horizontal Rafter	80x80x4 SHS	134	48.9%	✓ ADEQUATE
Kicker Braces	65x65x4 SHS	147	53.5%	✓ ADEQUATE

DESIGN LOADS

APPLIED LOADS:
- **Dead Load: 0.5 kN/m²**
- **Live Load: 0.6 kN/m²**
- **Total: 1.1 kN/m²**

INTERNAL FORCES:
- **UDL: 8.25 kN/m**
- **Max Moment: 3.93 kNm**
- **Max Shear: 8.06 kN**
- **Wall Reaction: 8.06 kN**

MATERIAL:
- **Steel Grade: S275**
- **Yield Strength: 275 N/mm²**

MATERIAL SCHEDULE & QUANTITIES

Member	Size	Length(mm)	Qty	Total(m)	kg/m	Weight(kg)
Wall Plate	50x50x3 SHS	7500	1	7.500	4.57	34.3
Spine Beam	50x50x3 SHS	7500	1	7.500	4.57	34.3
Horizontal Rafter	80x80x4 SHS	977	1	0.977	9.63	9.4
Kicker Braces	65x65x4 SHS	977	5	4.883	7.65	37.4

TOTAL LENGTH: 20.860 m | TOTAL WEIGHT: 115.3 kg

COST ESTIMATE

COST BREAKDOWN:

Steel Fabrication:
115.3 kg × $8.50/kg
= $980

Hot Dip Galvanizing:
115.3 kg × $2.20/kg
= $254

TOTAL MATERIAL: $1234

Excludes:
- **Installation labor**
- **Wall fixings & anchors**
- **Roofing/cladding**
- **Engineering certification**

FABRICATION NOTES & SPECIFICATIONS

MATERIALS & STANDARDS:
• All structural steel to AS/NZS 3679.1 Grade 300 minimum • Welding electrodes to AS/NZS 2717.1 • Bolts & fasteners Grade 8.8 minimum
• Hot dip galvanizing to AS/NZS 4680 • All welding to AS/NZS 1554.1 GP • Surface preparation to AS/NZS 1627.4

FABRICATION REQUIREMENTS:
• All connections minimum 6mm fillet welds • Main rafter connections full strength butt welds • Back gouge and re-weld critical connections
• NDT testing if specified by engineer • Straightness tolerance L/500 maximum • Overall dimensional tolerance ±5mm

INSTALLATION NOTES:
• Fix wall plate with M16 Grade 8.8 expansion anchors @ 400mm c/c maximum • Minimum edge distance 100mm from masonry edges
• Provide 8mm backing plates where wall condition requires • Check wall capacity for 8.06 kN reaction force
• All field connections to be welded by certified welder • Touch up galvanizing after installation welding

QUALITY ASSURANCE:
• Visual inspection of all welds • Dimensional check before galvanizing • Final inspection after installation
• Provide test certificates for all materials • Welding procedure specifications available • Structural engineer certification required